January 20, 2026

AAE 438
Airbreathing Propulsion
Spring 2026

Registration

This course has two sections, with each having an equivalent three credit hour value. The meeting time and delivery mode is as follows:

CRN 12476: In-person, on-campus
CRN 32576: In-person, on-campus

Instructor Information

Prof. Suman Chakraborty (*pronounced 'soo - man'*)
Office:
Email:
Office Hours:

Teaching Assistant Information

Alex Zhou
Email:
Office Hours: (Zoom Link)
(Zoom Link)
and by appointment

Tsean Mattoo
Email:
Office Hours: (Zoom Link)
(Zoom Link)
and by appointment

Course Information

Objectives

The objective of this course is to provide students with an introduction to air-breathing propulsion devices found in aerospace systems. The course will build from a foundation of thermodynamics and compressible flow to develop an understanding of engine performance and design. Technical coverage of major propulsion system elements will be provided, including inlets, compressors, combustors, turbines, afterburners, and nozzles. Modern trends and practices in engine design will also be emphasized. Upon completion of the course, students will have the ability to conduct preliminary design and predict the performance of air-breathing propulsion systems and their components.

Prerequisites

This course will build on previous undergraduate preparation in the thermal sciences, including thermodynamics, compressible aerodynamics, and heat transfer; specifically ME 200, AAE 334, and AAE 338.

Textbooks

The main textbook for this class will be:

Mattingly and Boyer, *Elements of Propulsion: Gas Turbines and Rockets*, 2016, ISBN 978-1-62410-371-1

The reference books below are all available through the Purdue Library (Aerospace Research Central) and serve as some general background material:

Hill and Peterson, *Mechanics and Thermodynamics of Propulsion*, 1992, ISBN 0-201-14659-2
Mattingly, Heiser, and Pratt, *Aircraft Engine Design*, 2002, ISBN 1-56347-538-3
Oates, *Aerothermodynamics of Gas Turbine and Rocket Propulsion*, 1998, ISBN 1-56347-241-4
Oates, *Aerothermodynamics of Aircraft Engine Components*, 1985, ISBN 0-915928-97-3

Websites

(1) For course administration, announcements, lecture notes, and grades:
Brightspace (all sections), *Spring 2026 AAE 438 - All Sections - Merge*

(2) For assignments:
Gradescope, Course ID 1211434, *AAE 438*

Course Notes

Structured notes to guide the lectures will be posted on Brightspace prior to the scheduled on-campus lecture.

Grading

Final grades will be assigned based on the homework assignments, two midterm exams, and a comprehensive final exam, with the following weights:

Homework Assignments:	20%
Midterm Exam 1:	25%
Midterm Exam 2:	25%
Review Report:	10%
Final Exam:	20%

A grade is assigned at the end of the semester based on a curved scale of all students in the class. Those who have demonstrated an excellent understanding of the material will be given an A+, A, or A- grade. Those who have demonstrated a minimum acceptable understanding will receive a C+, C, or C- grade, with B grades falling in between these conditions. Those who have demonstrated an unacceptable level of understanding will receive a D or F grade.

Homework

Approximately 7-8 homework assignments will be given at various times during the semester. The homework problems will be posted on the course website, and students are responsible for accessing them and submitting the solved problems (via Gradescope) by 11:59 PM ET on the noted due date. **A 24-hour extension is available; however, it comes with a 40% late submission penalty**. No homework will be accepted later than that, and all problems must be handed in at the same time. Exceptions will be granted only with the instructor's prior permission and only under special circumstances. **Moreover, your lowest homework grade will be dropped before the computation of your final grade**. The class is encouraged to work together on assignments and to seek help from the TAs or the Professor (if needed). However, the submitted assignment must be the individual student's work. The submission must also be legible and well-organized, or points may be deducted. *Good engineers must be able to communicate results. Homework is one way we practice this skill. Therefore, you should make sure that homework is neat and well organized.*

Exams

The midterm exams will be administered outside of class hours. The dates of the midterm exams are listed on the course schedule, but may be subject to change depending on pace of material coverage. The time and date of the final exam will be posted once that information is released from the university scheduling group. All students must take the exam at the scheduled time. With the exception of an emergency, failure to take the exam at the scheduled time will result in a grade of zero for that exam. If you have a foreknown schedule conflict, then contact the professor as soon as possible.

Use of Artificial Intelligence (AI) or Large Language Models (LLM)

The use of AI and LLM will increase over the next several years. Therefore, it seems prudent to allow students to use AI while in school when appropriate. However, we must balance the use of AI with the desire for you to obtain knowledge and have the ability to identify when AI has "hallucinated". Finally, we acknowledge that copyright should be upheld and that we should not share the work of others with AI without their permission. With these ideas in mind, the following policies will be enforced in this class:

- You may use AI on Homework but not on the Exams. This policy aims to strike a balance between learning how to utilize AI and acquiring knowledge. Therefore, you are allowed to use AI while you are developing your knowledge in the field (on formative assessments), but not when your personal knowledge is assessed (on summative assessments)
- AI use must be tracked and acknowledged. If you used Generative AI programs such as ChatGPT or CoPilot to assist with your assignment, you must acknowledge its use by following the guidelines provided in Monash University's Acknowledging the Use of Generative Artificial Intelligence. Specify how and where your readers can expect to see the impact, and include an Appendix for the assignment that shows what aspects were generated. ChatGPT now includes the ability to share links to conversations, you can use extensions such as ShareGPT to share your ChatGPT conversations, or you can include screenshots. (This paragraph was adopted from Dr. Kirsten Davis and their ENE 50101 course syllabus)
- You may not share course materials with AI. Class videos, notes, and handouts are considered to be 'derivative works' of the instructor's presentations and materials, and they are thus subject to the instructor's copyright. As such, they should not be uploaded to any share site, sold or bartered without express written permission. To be clear: this means that uploading any course material on "CourseHero", ChatGPT, Copilot (even the Purdue-branded version), or similar websites is a violation of the syllabus and a violation of copyrights

Other Notes and Policies

Questions

You are encouraged to attend office hours as your first source of assistance in the course. The TAs and the instructor will host several offerings throughout the week. Additional assistance will also be provided by the TAs via email. Include *AAE 438* as the start of your email subject line to ensure emails are identified and prioritized. Monitor Brightspace closely for announcements of general interest. Email with the instructor should be restricted to communication regarding individual matters course administration and similar concerns.

Changes and Emergencies

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted on the course website or can be obtained by contacting the instructor via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Copyright

Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Attendance

In-person participation is strongly encouraged. The Purdue handbook sets the following general expectation:
"The University expects that students will attend classes for which they are registered. At times, however, either anticipated or unanticipated absences can occur. The student bears the responsibility of informing the instructor in a timely fashion, when possible. The instructor bears the responsibility of trying to accommodate the student either by excusing the student or allowing the student to make up work, when possible. The University expects both students and their instructors to approach problems with class attendance in a manner that is reasonable."

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Accessibility and Accommodations

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us know so that we can discuss options. We also encourage you to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health

If you find yourself beginning to feel some stress, anxiety and/or feeling a bit overwhelmed, try WellTrack, https://purdue.welltrack.com/. Sign in and find information and tools at your fingertips, available to you at any time. If you need support and information about options and resources, please see the Office of the Dean of Students, http://www.purdue.edu/odos, for drop-in hours (M-F, 8 am - 5 pm). If you are struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Purdue Honor Pledge

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."
https://www.purdue.edu/provost/teachinglearning/honor-pledge.html